CoRe-TFM: Provenance, Clean Replication and Deposit Readiness

Technical status report - 20 October 2026. Applies to the enhanced Elsevier research draft and the frozen 160-configuration combined evidence ZIP. This is an artifact-quality and reproducibility assessment, not a claim of journal acceptance or independent rerunning of the historical experiments.

Verified inputs

Check	Independent finding
Combined source archive	ZIP CRC valid; 414/414 SHA-256 entries verified.
Model results	160 archived NPZ prediction caches and JSON sidecars.
Held-out partition specifications	40 frozen randomized-split JSON records.
Historical model checkpoint fingerprints	160/160 saved sidecars have no authenticated model-weight SHA-256.
Processed datasets	Five pickle snapshots present and hashed; no original raw OpenML byte provenance in the evidence.

Four requested publication-preparation tasks

Task	What is ready now	External dependency
Checkpoint provenance	v8 GPU notebook prepares explicit checkpoint path/hash capture for fresh runs.	Historical loaded weights cannot be reconstructed from old prediction caches.
OpenML provenance	11 numeric dataset-ID lookup records are prepared (wine has two sources), with version and original-file hashing code.	Network run needed for publisher metadata/raw checksums; the complete historic preprocessing environment is unavailable.
Pinned clean replication	Six matched TFM jobs defined for Car, Nursery and Diamonds, with frozen label/split equality gates and score comparison.	Needs Colab GPU, new checkpoint downloads, authentication and execution.
Immutable deposit and authors	Checksummed S2 publication-candidate ZIP; metadata and sign-off checklist prepared.	Authors must approve rights/declarations and publish the record to obtain a DOI.

What the completed benchmark shows

At the predefined 256-row training context and 52-label selection budget, hard arithmetic-reference reconciliation reduced internal marginal inconsistency on all 80 matched primary comparisons but worsened held-out NLL in 67/80. This is distinct from proving that the proposed guarded gate improves performance. The all-dataset multiple-comparison-adjusted exploratory tests remain above the nominal 0.05 threshold.

Checkpoint and dataset provenance

Historical model weights: no retroactive attestation

The original sidecars identify **TabICLv2 2.2.0** and **TabPFN package 9.2.0 with a V3 factory**. They do not include a verified SHA-256 of the actual weight file each historic GPU process loaded. Model package versions, checkpoint filenames, and hashes of prediction NPZ files are not sufficient to prove loaded checkpoint-byte identity.

The public TabICLv2 checkpoint listing identifies **tabicl-classifier-v2-20260212.ckpt**, with published SHA-256 `bdc7dbd5e4ff21f8f0456fcf90c6b7cdf72dbea960f2d05b19bec19f9b3d4ed0`. The new notebook treats this as a reference expectation, not historical proof. Official source: huggingface.co/jingang/TabICL.

For new TabPFN-3 runs, **ModelVersion.V3** must be explicitly selected because the default model version can change. The local checkpoint path selected by the factory is hashed after successful inference; a new-run path configuration and hash still need documented execution provenance. The V3 weight license is separate and noncommercial.

OpenML: distinguish dataset identity from downloaded bytes

The archived sidecars retain numeric OpenML IDs for ten paired-label datasets. One study dataset (Wine) combines two original dataset IDs; 11 separate source records therefore need validation. A source ID is reproducible metadata but not an original-file SHA-256, and processed pandas pickles are not equivalent to the original ARFF/Parquet bytes.

The v8 notebook fetches dataset metadata by numeric ID, records version, license, file identifier and publisher MD5, streams the supplied original download for SHA-256 and MD5 comparison, and refuses to silently claim verification when a download or checksum fails. Online checks were not executed in this CPU-only environment.

Data integrity gate before inference

The six replication configurations use the saved seed-23 / split-0 specification and 256-row training cap. New data must reconstruct the same class domains, the same ordered validation/test IDs, and exactly the same encoded target labels as the corresponding frozen cache. Any mismatch blocks inference or is reported as a failure. No old prediction NPZ is placed inside the fresh replication-results folder.

Environment record

The notebook requires the original TabICLv2 2.2.0 and TabPFN 9.2.0 package versions, and saves installed package versions, Python/PyTorch/CUDA/GPU driver details, the source-engine SHA-256, checkpoint path fingerprints and per-job elapsed time. A new environment manifest cannot retroactively identify the historic Colab environment precisely.

Data publication and sign-off

Prepared: S2 deposit candidate, not published

The prepared publication-candidate ZIP includes 160 model-prediction files and their sidecars, 40 frozen splits, independent audit tables, checksum manifests, source methods, reproducibility notebook, and deposit metadata. It excludes the five processed dataset pickles, raw dataset downloads, model weights, API tokens and other credential files. Every file in the candidate is protected by a new deposit manifest verified during packaging.

Scope limit: It represents October 19-20 evidence (S2). The earlier 12 October historical five-seed/candidate study (S1) is not present as verified, independent original notebook artifacts; it cannot be labeled as contained in S2.

Publication gate: must be completed by people

Gate	Required confirmation
Authorship and CRediT	Badampudi Agasthya Anirudh, Harshita Bogineni and Sri Preethaa K R approve names, roles and contributions.
Declarations	Authors confirm funding, conflicts, data usage, correspondence and publisher-required AI-use statement.
Rights	Verify underlying OpenML dataset rights and any restrictions on publishing outputs derived from TabPFN-3 noncommercial model weights.
Historical S1	Either locate and audit historic study artifacts or explicitly describe them as separately sourced evidence.
Repository and DOI	After author approval, publish supplementary files via a trusted repository such as Zenodo; reserve/publish a genuine DOI, then update the manuscript data availability section.

Where files belong

Required replication input: CoReTFM_FULL160_Combined_Verified_Evidence.zip (existing verified source).
New-run notebook: CoReTFM_v8_Clean_Replication_and_Provenance_No_Drive.ipynb.
Deposit candidate: CoReTFM_S2_Publication_Candidate_UNPUBLISHED.zip.
Sign-off: COAUTHOR_APPROVAL_AND_PUBLICATION_GATE.md.
Audit status: STATIC_PROVENANCE_AUDIT.json and DATASET_PROVENANCE_MATRIX.csv.

Publication judgment

The manuscript may be circulated as an enhanced **author-review draft**. The 160-cache audit is technically reliable for its saved predictions and derived scores. The phrase "fully reproducible checkpoint-verified study" remains unjustified until new GPU replication, raw dataset provenance, third-party rights checks and author sign-off have been completed.